Aarav Aggarwal

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EDUCATION

Singapore American School, Singapore *Aug 2021 - Present*

- Grade 9 GPA: 4.18; Grade 10 GPA: 4.29; Grade 11 GPA: 4.50 (Cum GPA: 4.32)

STANDARDIZED TESTS

- **SAT**: 1570/1600 – Math 790, Reading 780
- **APs**: Scored 5 in all 8 APs; pursuing 2 more this year
    - AP Computer Science, AP Calculus BC, AP Physics 1, AP Chemistry, AP Microeconomics, AP Macroeconomics, AP African American Studies, AP Psychology
    - *Pursuing:* AP Physics C Electromagnetism, AP Physics C Mechanics

HONORS & AWARDS

- Professional Cricket: Represented Singapore in U16 National team at Asian Cricket Council Tournament in 2025, finished Runner-Up (10 countries)
- Selected for Social Science (top 2% students) and Science (top 3% students) Honours Society
- Grade 9, Grade 10 and Grade 11 Scholars List: Awarded to top performing students
- Featured in Economic Times (Tech)—world's 2nd most-read English business newspaper—for Roof of Hope
- Published Author, Elaine Hobson Essay Competition: 500-word essay, "Ripple Effect" (July 2024)

INNOVATION AND PUBLICATIONS

Roof of Hope | Founder (https://www.roofofhope.org) *Dec 2024 - Present*

- Installed UV reflective low-cost tarpaulin cool roofs in 100 low-income homes as a pilot in summer 2025; Scaled to 1000 homes in summer 2026 through $40,000 in funding from Nagarro Foundation
- Collaboration with Mahila Housing Trust and Indorama Ventures for Clean Energy
- Findings reveal a significant reduction in over deck (15–20 °C) and indoor ceiling temperatures (4–11 °C), household surveys confirm substantial improvements in thermal comfort, sleep quality
- Research publication accepted to 7th International Conference on Counter Measures to Urban Heat Islands, 2026

Bio-Electrochemical Cells | Independent Research *July 2025 - Present*

- Designed a bio-electrochemical cell with aloe vera and orange peel as sustainable electrolytes
- Authored findings showing aloe vera sustaining 1.393 V mean output across a 7-day trial
- Published in Oxford Journal of Student Scholarship (Vol. 11, No. 5, Sep 2026)
- Secured 2 Design Patents, Indian Patent Office—low-cost eco-friendly plant-based battery design

DataPitch | Developer *Oct 2024 - Present*

- Built an ML model to predict player performance in the Indian Premier Cricket League (IPL) using K-Means clustering, PCA, and Linear Regression quantifying the link between batting role composition and team win rate
- Published in National High School Journal of Science (NHSJS Reports: April 23, 2026)
- Won CREST Gold Award (by British Science Association)

OTHER ENGINEERING & TECHNOLOGY EXPERIENCE

M&TSI | Jerome Fisher Management & Technology Summer Institute, Univ of Pennsylvania *July 2026*

- 3 week highly selective summer program, ~75 students globally to explore intersection of business and technology
- College credited course comprising engineering labs, entrepreneurship, financial modelling, business simulations
- Built a working prototype using a two-axis active stabilization platform for use in moving cars with an inertial measurement unit, microcontroller, servos, custom-designed parts, and PID-control code

Young Technology Scholars | Plaksha University *Jun 2023*

- Selective summer program for high school students exploring intersection of tech and real-world applications
- Built a prototype of Remote Surgery Teleoperation Bot; with ESP-32, gyroscope for a hand-controlled robot

INTERNSHIPS

Mynzo Carbon | Machine Learning Intern *Jun - Aug 2025*

- Developed a leaf-species identification pipeline from in-situ photos using MobileNetV2 transfer learning, robust augmentation, and cross-validated evaluation
- Curated a 250-image dataset across 10 species and achieved ~85% accuracy; applied the model to tag field samples, accelerating species-distribution assessment for high-integrity carbon projects

Council on Energy, Environment and Water | Research and Policy Intern *Jun 2026*

- Analyzed 150+ heat action plans to identify possible interventions and eligibility for government funding
- Designed vulnerability surveys including household questionnaires and for data collection on heat exposure

TEACHING & MENTORING

Plaksha University | Teaching Assistant *Jun 2024*

- Invited to be a Teaching Assistant at Young Tech Scholars program
- Assisted 80+ students on a Teleoperation project through live lessons, instructional videos and hands-on demonstrations

Read to Lead | Vice President, Singapore American School *Aug 2024 - Present*

- Partnered with the Singapore National Library Board to launch monthly literacy workshops, reaching 100+ young readers through interactive sessions; expanded program across 4 Asian countries
- Co-designed curriculum and initiated a book drive to provide underprivileged children with access to 500+ books
- Expanded program to 4 Asian countries

EXTRACURRICULARS/ COMMUNITY INITIATIVES

Singapore Cricket Association | Singapore U16 National Team & U19 High-Performance Squad *Jan 2018 – Present*

- Selected to represent Singapore U16 National Cricket Team at the Asian Cricket Council (ACC) U16 Championship; achieved Runner-Up finish in 10 country tournament
- Played competitively for 8 years, train 5 days weekly with National High-Performance Squad, SCA Men's League

Cricket Club | Officer, Singapore American School. *May 2024 - Present*

- Led 5+ inter-school events and 2 international tours (Philippines, Malaysia) to grow cricket participation
- Raised funds and implemented structured training upgrades, including permanent nets and a development program benefiting 40+ student athletes

Stumps and Stories | President, Singapore American School *Oct 2024 - Present*

- Organized 2 school wide events honoring over 160+ migrant workers and staff, fostering community appreciation and inclusivity through sport and storytelling
- Designed awareness campaigns and fundraised a $5,000 to support event operations and commemorative kits

Cool Play | Founder *May - Aug 2025*

- Ran a fundraising campaign raising $1,000+ to distribute 60+ cooling jackets to young athletes, resulting in their improved performance, reduced incidents of heat exhaustion